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Unit “ 5 : Session “ 4 : SLO “ 1 | Parallelism, Control Unit and Pipelining Review

■ I. Multiple Choice

1. b) Executing multiple instructions simultaneously within a single processor.
2. c) Coordinate and control the operations of other units.
3. c) Overlapping multiple instruction stages.
4. c) Synchronization and communication overhead.
5. b) Memory consistency problems “ shared data cached per-core must stay coherent/consistent.

■ II. True or False

1. FALSE “ Amdahl's Law: the serial fraction and overheads prevent linear speedup.
2. TRUE.
3. TRUE.
4. FALSE “ communication overhead increases with more processors.
5. TRUE.

■ III. Scenario-Based Activity

1. Idle/overloaded cores = load imbalance “ solution: dynamic work scheduling (work stealing, finer-grained tasks).
2. Poor scaling with more cores = Amdahl/serial fraction + synchronization overhead “ solution: reduce the serial portion, minimize lock scope, use lock-free structures.
3. Results differ between runs = memory consistency/race conditions “ solution: proper synchronization (locks, atomics, barriers) or a defined consistency discipline.
4. Waiting for data updates = communication overhead “ solution: data locality, partition work to minimize sharing, overlap communication with computation.